

Question 5 – Prime Palindrome

Date: 24/07/2026

Objective / Aim

To check whether a given number is both prime and palindrome using Java.

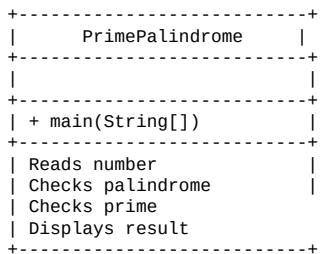
Algorithm / Logic

1. Read an integer from the user.
2. Store the original number in another variable.
3. Reverse the number using repeated remainder and division operations.
4. Compare the reversed number with the original number to check palindrome.
5. Check whether the number is prime by testing divisibility from 2 to $x/2$.
6. If both prime and palindrome conditions are true, display PrimePalindrome.
7. Otherwise, display Not PrimePalindrome.

Steps for Execution

1. Create the Java project and add PrimePalindrome.java.
2. Compile and run the program.
3. Enter an integer.
4. The program reverses the number and checks palindrome.
5. The program checks whether the number is prime.
6. Observe the final result.

UML Class Diagram



Source Code

```
import java.util.Scanner;

public class PrimePalindrome{
    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        System.out.println("Enter a Number : ");
        int x = s.nextInt();
        int y = x;
        int r=0;

        while(y>0){
            r=(r*10)+(y%10);
            y=y/10;
        }

        boolean prime=true;
        boolean palin=false;

        if(r==x)
            palin=true;

        for(int i=2;i<=x/2;i++){
            if (x%i==0){
                prime=false;
                break;
            }
        }

        if(prime && palin)
            System.out.println("Given number is a PrimePalindrome");
        else
            System.out.println("Given number is Not PrimePalindrome");
    }
}
```

Output

```
Enter a Number :
131
Given number is a PrimePalindrome
```